SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct

I JAGADISH.V(192521352)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: jagadish

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1. Cloud Architecture for Video Streaming Platform

Architecture Design

- Use Content Delivery Network (CDN) to cache videos closer to users.
- Deploy Load Balancer to distribute traffic across multiple application servers.
- Store videos in cloud object storage.
- Use Auto Scaling Groups to automatically add or remove servers based on demand.
- Use Database Replication for high availability.

Load Balancing and Auto-Scaling

- Load Balancer evenly distributes user requests, preventing server overload.
- Auto-Scaling automatically launches new instances during peak traffic and removes them during low traffic.
- This ensures smooth video streaming with minimal buffering.

Cost Optimization Strategies

- Use pay-as-you-go pricing.
- Store frequently accessed videos in CDN cache.
- Move old videos to low-cost archival storage.
- Use reserved instances for predictable workloads.
- Monitor resource usage and remove idle resources.

Conclusion

A scalable cloud architecture with CDN, load balancing, and auto-scaling ensures high performance, low buffering, and reduced operational cost.

2. Multi-Cloud Container Deployment for FinTech Startup

Proposed Solution

- Package applications using Docker containers.
- Use Kubernetes for container orchestration across multiple cloud providers.
- Deploy workloads across AWS, Azure, and Google Cloud for high availability.

Service Discovery

- Kubernetes provides DNS-based service discovery, allowing containers to communicate automatically.
- Services can locate each other without manual configuration.

Scaling

- Horizontal Pod Autoscaler (HPA) increases or decreases container replicas based on CPU or memory usage.
- Cluster Autoscaler adds or removes worker nodes automatically.

Benefits of Multi-Cloud

- High availability
- Disaster recovery
- Avoids vendor lock-in
- Better global performance

Risks

- Increased management complexity
- Higher networking costs
- Security policy differences
- Data synchronization challenges

Conclusion

Kubernetes with a multi-cloud strategy provides consistent deployments, automatic scaling, and improved reliability.

3. Backup and Disaster Recovery for E-Learning Platform

Backup Strategy

- Schedule automatic cloud-native backups.
- Enable versioning to recover deleted or modified files.
- Replicate data to another cloud region.
- Take periodic database snapshots.

Achieving RTO and RPO

Recovery Time Objective (RTO)

- Use standby servers and automated failover.
- Fast recovery minimizes downtime.

Recovery Point Objective (RPO)

- Enable continuous data replication.
- Frequent backups reduce data loss.

Business Continuity Steps

1. Enable automated backups.
2. Replicate data to another region.
3. Configure disaster recovery site.
4. Test backup restoration regularly.
5. Monitor backup health.
6. Document disaster recovery procedures.

Conclusion

Cloud-native backups, replication, and versioning ensure quick recovery and minimal data loss during failures.

4. Hybrid Cloud Architecture for Government Agency

Hybrid Architecture Design

- Store sensitive data on-premises.
- Store non-sensitive applications in the public cloud.
- Connect both environments using VPN or dedicated private connection.
- Secure communication through encrypted channels.

Identity Management

- Implement Single Sign-On (SSO).
- Use Identity Federation between on-premises and cloud.
- Apply Role-Based Access Control (RBAC).
- Enable Multi-Factor Authentication (MFA).

Encryption

- Encrypt data at rest using AES-256.
- Encrypt data in transit using TLS/SSL.
- Secure encryption keys with cloud key management services.

Challenges

- Complex infrastructure management
- Compliance requirements
- Network latency
- Security monitoring across environments
- Integration between cloud and on-premises systems

Conclusion

A hybrid cloud provides flexibility while maintaining security and regulatory compliance for sensitive government data.

5. High Availability and Fault Tolerance for SaaS Provider

Multi-Region Deployment Strategy

- Deploy application servers in multiple cloud regions.
- Use a Global Load Balancer to route users to the nearest healthy region.
- Maintain identical infrastructure in all regions.

Data Replication

- Configure synchronous replication for critical databases.

- Use asynchronous replication for disaster recovery.
- Replicate storage across regions.

Failover Mechanism

- Continuously monitor application health.
- Automatically redirect traffic to another region if one fails.
- Use DNS failover and health checks.

Monitoring and Alerting

- Monitor CPU, memory, storage, and network usage.
- Track application response time and error rates.
- Configure alerts for server failures, high latency, and resource exhaustion.
- Use dashboards and automated notifications for administrators.

Conclusion

A multi-region architecture with replication, automatic failover, and continuous monitoring ensures high availability and uninterrupted service even during regional outages.